

Krish Chawla

Work Eligibility: Curricular Practical Training (CPT) - Student (F-1) Visa
krishschawla@gmail.com | Fredonia, NY | 716-732-4307

EDUCATION

Bachelor of Science, State University of New York at Fredonia, expected December 2026

Major: Computer Science | Dual Minors: Applied Mathematics and Data Science | GPA: 3.92

TestOut Security Pro Certification, CompTIA, December 2024

Computer Science Department, SUNY Fredonia

INTERNSHIPS & RELATED EXPERIENCE

Artificial Intelligence Intern, SUNY Fredonia Center for Innovation and Economic Development, Dunkirk, NY (Jan 2026–Present)

- Planned work on prompt engineering and AI pre-trained model deployment

Robotics Project, Computer Science Department, SUNY Fredonia (Sep 2024–Dec 2024)

- Configured next-generation robots in the Robotics Lab under faculty guidance

Traffic Analytics Data Programming Intern, SUNY Fredonia Center for Innovation and Economic Development, Dunkirk, NY (Jun–Aug 2024)

- Analyzed traffic data to determine vehicle usage and predict congestion
- Used Linux, Python, YOLO libraries, and Supervision for traffic analysis
- Optimized store counts in databases and visualized results using line charts
- Recorded average vehicle speed per minute and noted special traffic conditions

PROJECTS

Jirassic Park — Agile Task Management Platform | React, TypeScript, Express, PostgreSQL, Prisma, Auth0 (Spring 2026)

Team project, CSIT 425: Software Engineering — github.com/burk3405/jirassic-park

- Built backend REST API for task CRUD, sharing (VIEW/EDIT permissions), and advanced search/filtering
- Implemented Auth0 JWT authentication middleware and CORS configuration; debugged a Prisma ORM WHERE-clause precedence bug affecting task ownership filtering
- Wrote unit test suite achieving 27/27 passing tests

- Applied the Repository design pattern via Prisma ORM for type-safe, SQL-free data access

Rubik's Cube Solver — Optimal Search Algorithm | C/C++ (individual project) CSIT 441: Analysis/Design of Algorithms

- Implemented Korf's Algorithm (IDA* search with Pattern Database heuristics) to solve the Rubik's Cube optimally across its $\sim 4.3 \times 10^{19}$ -state space
- Built pattern database heuristic tables and a benchmarking harness to empirically validate solve time and node expansion against theoretical complexity bounds
- Confirmed pattern-database-guided search drastically reduced nodes expanded relative to naive breadth-first search, consistent with theoretical expectations

Non-Preemptive CPU Scheduling Simulation | Python (Spring 2026)

Team project, CSIT 431: Intro to Operating Systems

- Implemented FCFS, SJF, and Priority Scheduling algorithms in an object-oriented, tick-based simulation
- Ran controlled experiments (10 trials $\times$ 15 processes) comparing algorithm performance on waiting time and turnaround time
- Found Priority Scheduling outperformed FCFS by $\sim 37\%$ in average waiting time (136.7 vs. 217.7 ticks)

Predicting UEFA Champions League Success | Python, scikit-learn (Spring 2026)

STAT 405: Statistics Seminar — Presented at SURC West Research Conference, April 2026

- Engineered per-90 performance metrics from 6 merged Kaggle/UEFA 2021–22 datasets covering goals, attacking, defending, and goalkeeping
- Applied K-Means clustering to identify 4 player archetypes and Random Forest (5-fold CV) to predict match outcomes from team-level stats
- Found goal-scoring output as the strongest predictor of match success, with pass accuracy as a multiplying factor

RESEARCH

Undergraduate Researcher — Fighter Aging Curves in the UFC, SUNY Fredonia, Faculty Sponsor: Dr. Megan Johnson (Fall 2026–Spring 2027)

- Recipient of the 2026 Donald Nasca Undergraduate Student Research and Creativity Award (\$2,500, fully funded)
- Building empirical aging-curve models for UFC fighter performance using Python (pandas, BeautifulSoup) and R, applying delta-method and mixed-effects regression to career-long striking, takedown, and win-rate data
- Selected for presentation at the 2027 OSCAR Student Research and Creativity Exposition

Team Member, 2026 Mathematical Contest in Modeling (MCM), COMAP, Faculty Advisor: Lan Cheng, SUNY Fredonia (2026)

- Solved MCM Problem C ("Data With The Stars") with a 3-person team, modeling unknown fan vote estimates from Dancing with the Stars judges' scores and contestant data across 34 seasons
- Compared rank-based and percent-based vote-combination methods to evaluate fairness of the show's elimination system
- Designated a Successful Participant

LEADERSHIP

Peer Tutor — Computer Science & Mathematics, SUNY Fredonia (Aug 2025–Present)

- Completed EDU-100-01: Tutoring Theory & Practice certification training
- Tutor 100+ students weekly across Computer Science and Mathematics courses
- Submit three formal tutoring reports per semester and attend monthly check-ins with tutoring coordinator

Computer Lab Proctor, Computer Science Department, SUNY Fredonia (Feb 2024–Present)

- Monitor student access to the computer lab
- Troubleshoot technical problems such as printing or software errors

Programming Languages Teaching Assistant, SUNY Fredonia (Feb 2024–Present)

- Support student learning with tutoring sessions and office hours
- Verify accuracy of classwork and homework assignments

LICENSES & CERTIFICATIONS

How IBM Bob and AI Tools Are Changing the Way Solutions Are Built, IBM (Issued Jul 2026)

CPR/AED/First Aid, Athletics Department, SUNY Fredonia (Issued Aug 2024)

Dean's Honors List, State University of New York at Fredonia (Issued Aug 2023 to present)

- Consistently named to the Dean's List every semester since freshman year

TECHNICAL SKILLS

Languages: Python, C, C++, R, TypeScript, React.js, Prisma, PostgreSQL

Software: Linux, GitHub, VS Code, IntelliJ, Neovim, Docker, AWS, Azure

Analytics & Tools: Apache, CorelDraw, Auth0, HuggingFace, YOLO, SQL, GCP

STUDENT ENGAGEMENT & CULTURAL PARTICIPATION

- Chess Club, SUNY Fredonia (Aug 2023–Present)
- Commencement Event Volunteer, SUNY Fredonia (May 2024)
- Student Presenter Outreach & Admissions, International Student Services (Apr 2024)
- Multicultural Weekend Visitor Host – Hosted 4 High School students (Mar 2024)

GENERAL EMPLOYMENT

- Fitness Center Desk Attendant, Blue Devils Fitness Center, SUNY Fredonia (Aug 2024–Present)
- Desk Attendant (Substitute), E-Sports Lounge, SUNY Fredonia (Oct 2023–Present)
- Food Service Attendant, Cranston Marche Dining Hall, SUNY Fredonia (Jun–Aug 2024)

AWARDS & ACCOMPLISHMENTS

- Alpha Lambda Delta Honor Society Member (Inducted Mar 2025)
- Pi Mu Epsilon National Mathematics Honor Society (Inducted 2026)
- Highest Commendation Certificate, United Nations Summit on Mental Health (2020)